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Subject : DA and AI

Question 1: What is Artificial Intelligence?

ANS.

AI can be defined as the branch of computer science that can simulate and perform human intelligence. AI is implemented in machines that can perform tasks that requires human intelligence and mind. Some of the primary function include Reasoning, Problem solving, learning, quick decision making in AI.

At the core of it, Artificial intelligence is nothing but algorithms with a certain set of rules. AI system have ability to learn from iteration of tasks where computer data (Also known as set of algorithms) are fed to machine. This is

how machine learning can get better in their specific tasks, without external interference .

Artificial intelligence can be classified into 3 parts:

1. Artificial Narrow intelligence (Narrow or Weak AI)

2. Artificial General intelligence

3. Artificial Super intelligence

Question 2: What is data analytics. Explain various steps involved in it.

ANS. In a simple way data analytics is Science of analysing raw data and to make conclusion about that information.

There are six steps in data analytics:

- The first step in the process is to Ask. The data analyst is given a problem/business task.
- The second step is to Prepare or Collect the Data. • The third step is Data Process and Clean
- The fourth step is to Analyse.
- The fifth step is Share.
- The final/sixth step is Act or Report about the data.

All these steps are necessary in data analytics. In the first step, We ask about the information of data or about the data itself. The user give us the asked data . after that second step is to collect the data given by the user. We collect the data or We prepare the data according to the given information by the user. Third step is to Process and clean the data. We make and adjust the data according to our need and also remove unnecessary data from it. In the fourth step, we simply analyse data and collect necessary information from it. The fifth step is to share the data and information which we gathered while analysing it. The last step is to Act or give the report about the entire data to the user itself.

Question 3. Explain comments in python .

ANS.

In Python, comments are descriptions that help programmers to understand the functionality of the program. Thus, comments are necessary while writing code in Python.

In Python, we **use the hash (#) symbol to start writing a comment**.

The comment begins with a hash sign (#) and whitespace character and continues to the end of the line.

In a team, many programmers work together on a single application. Therefore, if you are developing new code or modifying the existing application's code, it is essential to describe the purpose behind your code using comments.

For example, if you have added new functions or classes in the source code, describe them. It helps other colleagues understand the purpose of your code and also helps in **code review**.

Apart from this, In the future, it helps to **find and fix the errors**, **improve the code later on**, and reuse it in many different applications.

Writing comments is considered a good practice and required for **code quality**.

If a comment is found while executing a script, the Python interpreter completely ignores it and moves to the next line.

Two types of comments in Python:

1. Single line comment (# before starting your comment)
2. Multi line comment(use # before every commenting line)

Question 4: What is keywords and identifiers in python.

ANS.

Python keywords are the python reserved words reserved by python language secured to define the python syntax and structure of the python language. We cannot use these reserved keywords as variable names, function names, class names, or other identifiers. In python, all the keywords are case-sensitive. All the python keywords should contain lowercase letters only except True, False, and None must be written as they are.

Some of python keywords:

Keyword	**Description**
True	This keyword represents the Boolean Value 'true.'
False	This keyword represents the Boolean Value 'false.'
None	This keyword represents the Null Value
and	This is a logical operator

A Python identifier is a name used to identify a variable, function, class, module or other object. An identifier starts with a letter A to Z or a to z or an underscore (_) followed by zero or more letters, underscores and digits (0 to 9).

Python does not allow punctuation characters such as @, $, and % within identifiers. Python is a case sensitive programming language. Thus, **Manpower** and **manpower** are two different identifiers in Python.

Here are naming conventions for Python identifiers –

- Class names start with an uppercase letter. All other identifiers start with a lowercase letter.
- Starting an identifier with a single leading underscore indicates that the identifier is private.
- Starting an identifier with two leading underscores indicates a strongly private identifier.
- If the identifier also ends with two trailing underscores, the identifier is a language-defined special name.

Question 5: Explain different techniques in data analytics.

ANS.

Data analytics techniques:

1. Cluster Analysis

It is a simple technique of classifying data into groups or categories known as clusters. A cluster analysis identifies structures within a given dataset. Thus, you will get multiple groups, with each group internally containing homogeneous data while being heterogeneous to each other externally.

2. Factor Analysis

It is a technique of condensing large datasets with multiple variables into fewer variables. For example, in a set of 50 variables, there may be some that are correlated on several bases. These relations are latent or hidden. Factor analysis brings out these correlations as factors, giving a reduced number of variables.

3. Regression Analysis

When you talk of analyzing numerical data, regression analysis is the most common technique. It is basically used to determine the relationship between a dependent (main) variable and multiple independent variables (factors impacting the dependent variable).

4. Time Series Analysis

You may often want to analyze data for a given period in order to understand the trend of that particular time. For example, weekly revenues, monthly customer subscriptions, and sign-ups, etc. Time series analysis is primarily conducted to forecast future trends and cycles.

5. Decision Tree Analysis

A decision tree analysis is used to make decisions and choose the most advantageous options for your business. It is a diagrammatic layout of the possible risks and rewards you will get with every decision you make.